

📖 AI/ML Internship – Day 23 Notes & Tasks

🟡 Module 4: NumPy (Broadcasting – Very Important Concept)

📌 Part 1: What is Broadcasting?

💠 Definition

👉 Broadcasting is a NumPy feature that allows arrays of different shapes to perform operations together automatically.

👉 NumPy automatically adjusts smaller arrays to match larger arrays.

📌 Part 2: Why Broadcasting is Important?

👉 Without broadcasting:

- We need loops
- Code becomes slow

👉 With broadcasting:

- Faster operations
 - Cleaner code
 - Efficient memory usage
-

📌 Part 3: Theory Answers (IMPORTANT)

💠 1. What is Broadcasting?

👉 Broadcasting is the automatic expansion of smaller arrays during operations with larger arrays.

💠 2. Why Broadcasting is Useful?

👉 It:

- Reduces loops
- Improves speed

- Makes code simpler
-

◆ 3. What Happens During Broadcasting?

👉 NumPy compares shapes:

- If compatible → operation happens
 - If incompatible → error occurs
-

◆ 4. Broadcasting Rule

👉 Two dimensions are compatible when:

- They are equal
OR
 - One of them is 1
-

◆ 5. Advantage of Broadcasting

👉 Faster mathematical operations on datasets.

📌 Part 4: Simple Broadcasting Example

```
import numpy as np
```

```
arr = np.array([1, 2, 3])
```

```
print(arr + 10)
```

◆ Output

```
[11 12 13]
```

👉 Here:

- 10 is automatically added to every element.
-

📌 Part 5: Array + Array Broadcasting

```
a = np.array([[1], [2], [3]])
```

```
b = np.array([10, 20, 30])
```

```
print(a + b)
```

◆ Output

```
[[11 21 31]
```

```
 [12 22 32]
```

```
 [13 23 33]]
```

📌 Part 6: Understanding Shapes

```
print(a.shape)
```

```
print(b.shape)
```

👉 Shapes:

```
(3,1)
```

```
(3,)
```

👉 NumPy expands arrays automatically.

📌 Part 7: Incompatible Shapes

```
a = np.array([1, 2, 3])
```

```
b = np.array([1, 2])
```

```
print(a + b)
```

👉 This gives error because shapes are incompatible.

📌 Part 8: Broadcasting in 2D Arrays

```
arr = np.array([
    [1, 2, 3],
    [4, 5, 6]
])
```

```
print(arr + 100)
```

👉 Output:

```
[[101 102 103]
 [104 105 106]]
```

📌 Part 9: Real-World Thinking

👉 In AI/ML:

- Add bias to neural network outputs
- Normalize datasets
- Scale image pixels

👉 Broadcasting helps perform:

- Large computations efficiently
 - Matrix operations quickly
-

📌 Part 10: Visualization of Broadcasting

Array A:

```
[[1]
 [2]
 [3]]
```

Shape → (3,1)

Array B:

```
[10 20 30]
```

Shape → (3,)

Broadcast Result:

[[11 21 31]

[12 22 32]

[13 23 33]]

Tasks for Day 23

Task 1: Theory (Write Answers)

1. What is broadcasting?
 2. Why broadcasting is important?
 3. What are broadcasting rules?
 4. What happens with incompatible shapes?
 5. Advantages of broadcasting
-

Task 2: Basic Broadcasting

1. Add scalar to array
 2. Multiply array with scalar
 3. Divide array by scalar
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Task 3: Array Broadcasting

1. Add two compatible arrays
 2. Multiply arrays
 3. Observe shape changes
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Task 4: Shape Practice

1. Print shapes
2. Check compatible shapes
3. Try incompatible shapes

 Task 5: 2D Broadcasting

1. Add scalar to matrix
2. Multiply matrix
3. Normalize matrix values